

Supplementary Material

Exploratory resting-state EEG associations with anxiety and perceived social support in two distinct samples

S1. Scope and inferential structure

This supplement organizes the auditable evidence supporting the manuscript. The participant, not the temporal window, was the inferential unit. It contains only aggregate or nonidentifying derived results; local raw EEG and individual-level psychometric data are not included.

The local sample comprised 25 participants and three EEG representations: eyes closed (EC), eyes open (EO), and EC–EO reactivity. The external analytical sample comprised 135 nonoverlapping LEMON participants; external moderation used variable-specific complete-case samples of 108–135.

S2. Why 51 selections produced 50 unique targets

The 51 rows were selected interaction-model rows, not 51 distinct channel–band targets. The target key used for external feature-level analysis was EEG representation + local outcome + EEG identifier. One target occurred in two local moderator models and was therefore counted twice among selection rows but once after target-level deduplication. The component was retained in both corresponding signatures; only the feature-level target list was deduplicated.

Table S1. The duplicated local selection that accounts for 51 rows but 50 unique targets.

Target	Representation	Outcome	Channel	Band	EEG ID	Local models	Consequence
TARGET__DELT A_OC_ MINUS_ OA__M SPSS__E EG23	DELTA_OC_MINUS_OA	MSPSS	C4	alpha	EEG23	W3 W4	Two selection rows; one transferred target; retained in both signatures

W3 represented sleep deviation and W4 represented living alone in the local models.

S3. Frozen signature specification and sample-specific standardization

The transported specification comprised the identities of the selected components and their directional equal-magnitude weights. These components and weights were frozen before the reported LEMON models. Centering and scaling parameters were not transported. Within each analytical sample, every component was standardized using that sample's mean and sample standard deviation (ddof = 1), after which the frozen weights were applied.

Table S2. Frozen signature definitions.

NET	Representation	Origin	Local W	k	Frozen EEG identifiers
NET01	OC	MSPSS	W4	31	EEG17 EEG18 EEG21 EEG22 EEG9 EEG10 EEG13 EEG14 EEG42 EEG45 EEG46 EEG1 EEG2 EEG5 EEG6 EEG33 EEG34 EEG37 EEG38 EEG25 EEG26 EEG29 EEG30 EEG57 EEG58 EEG61 EEG62 EEG49 EEG50 EEG53 EEG54
NET02	OA	MSPSS	W4	2	EEG61 EEG62
NET03	DELTA_OC_MI NUS_OA	MSPSS	W3	8	EEG19 EEG23 EEG11 EEG43 EEG39 EEG31 EEG59 EEG55
NET04	DELTA_OC_MI NUS_OA	MSPSS	W2	3	EEG45 EEG7 EEG51
NET05	DELTA_OC_MI NUS_OA	MSPSS	W4	1	EEG23
NET06	DELTA_OC_MI NUS_OA	BAI	W4	4	EEG12 EEG4 EEG8 EEG59
NET07	DELTA_OC_MI NUS_OA	BAI	W2	1	EEG3
NET08	DELTA_OC_MI NUS_OA	MSPSS	W5	1	EEG36

NET is a computational identifier and does not imply a functional, anatomical, or connectivity network.

S4. Complete permutation manifest

All empirical values used the finite-sample correction $p = (b + 1)/(B + 1)$, where b is the number of null statistics at least as extreme as the observed statistic and $B = 10,000$. Local correlation and selection-aware analyses used seed 20260726. External analyses used seed 20260727 and reused one subject-ranking scheme across the specified external families.

Table S3. Permutation analyses and multiplicity families.

ID	Family	N/tests	B/seed	Exchangeability scheme	Statistic	Multiplicity
P01	Bivariate EEG correlations	25/576	10000/20260726	One common 25-subject row permutation applied jointly to BAI, MSPSS, and discrepancy; evaluated against EC, EO, and EO-EC features	Absolute Pearson r	BH within 64-feature blocks, within 192-test outcome families, and global 576; BY global; maxT at 64, 192, and 576 levels
P02	Selection-aware moderation signatures	25/30	10000/20260726	Residuals permuted within $W=0/W=1$ strata	Absolute t for signature $\times$ W after repeating screening and signature construction	BH and BY across 30 possible signatures; maxT by representation and globally; omnibus maxT
P03	Signature $\times$ moderator interactions	108–135/112	10000/20260727	Common random subject ranks subset to each moderator-specific complete-case mask	Absolute t for $X\times W$	Global BH/BY and maxT across 112; BH within W and signature; maxT within W
P04	Channel–band target $\times$ moderator or interactions	108–135/700	10000/20260727	Common random subject ranks subset to each moderator-specific complete-case mask	Absolute t for $X\times W$	Global BH/BY and maxT across 700; BH within W and target; maxT within W
P05	Matched signature–outcome main effects	135/8	10000/20260727	One common 135-subject order applied to each matched outcome	Absolute Pearson r	BH and maxT across eight matched pairs
P06	Matched channel–band main effects	135/50	10000/20260727	One common 135-subject order applied to each matched outcome	Absolute Pearson r	BH and maxT across 50 transferred targets
P07	All eight signatures versus MSPSS	135/8	10000/20260727	Same 135-subject orders used for all eight signatures	Absolute Pearson r	BH and maxT across eight signatures versus MSPSS

FL = Freedman–Lane residual permutation; BH = Benjamini–Hochberg; BY = Benjamini–Yekutieli.

S5. Analysis-family audit

Table S4. Summary of local and external analysis families.

Family	Tests	Nominal	Permutation	HC3+perm	BH $q < .10$	maxT $< .05$	Interpretation
Local bivariate correlations	576	0	0	—	0	0	No local bivariate signal
Local channel–band moderation	1920	51	—	—	0	—	51 nominal; none survived FDR
Local spatial aggregation	750	17	—	—	0	—	No aggregated model survived FDR
Selection-aware signature combinations	30	8	1	—	0	0	No selection-aware signature survived
External signature $\times$ moderator	112	9	9	6	0	0	No corrected external moderation
External target $\times$ moderator	700	55	53	42	0	0	No corrected external moderation
External matched signature main effects	8	5	5	—	5	1	NET03 was the only maxT $< .05$ result
External matched target main effects	50	26	26	—	26	2	T8 and Fp2 alpha EC–EO survived maxT
Post hoc all signatures versus MSPSS	8	6	6	—	6	3	Post hoc; not confirmatory

Dashes indicate that the corresponding inferential quantity was not defined for that family.

Table S5. Matched external signature–outcome main effects.

NET	Representation	Outcome	N	r	p	p perm	q BH	p maxT
NET01	OC	MSPSS_total	135	-0.173	0.0443	0.0461	0.0738	0.2267
NET02	OA	MSPSS_total	135	-0.214	0.0127	0.0132	0.0528	0.0794
NET03	DELTA_OC_MIN US_OA	MSPSS_total	135	-0.231	0.0071	0.0065	0.0520	0.0448
NET04	DELTA_OC_MIN US_OA	MSPSS_total	135	0.146	0.0921	0.0923	0.1231	0.4058
NET05	DELTA_OC_MIN US_OA	MSPSS_total	135	0.178	0.0385	0.0396	0.0738	0.2052
NET06	DELTA_OC_MIN US_OA	STAI_Trait_A nxiety	135	0.087	0.3181	0.3166	0.3166	0.8714
NET07	DELTA_OC_MIN US_OA	STAI_Trait_A nxiety	135	-0.174	0.0436	0.0446	0.0738	0.2244
NET08	DELTA_OC_MIN US_OA	MSPSS_total	135	0.112	0.1942	0.1951	0.2229	0.6865

Multiplicity values correspond to the prespecified family of eight matched signature–outcome pairs.

Table S6. Post hoc analysis of all eight signatures against LEMON MSPSS.

NET	Representation	Status	N	r	95% CI	p perm	q BH	p maxT
NET01	OC	same_construct_external_main_effect	135	-0.173	[-0.333, -0.005]	0.0461	0.0615	0.2010
NET02	OA	same_construct_external_main_effect	135	-0.214	[-0.370, -0.047]	0.0132	0.0264	0.0740
NET03	DELTA_OC_MIN US_OA	same_construct_external_main_effect	135	-0.231	[-0.385, -0.064]	0.0065	0.0219	0.0424
NET04	DELTA_OC_MIN US_OA	same_construct_external_main_effect	135	0.146	[-0.024, 0.307]	0.0923	0.1055	0.3677
NET05	DELTA_OC_MIN US_OA	same_construct_external_main_effect	135	0.178	[0.010, 0.337]	0.0396	0.0615	0.1807
NET06	DELTA_OC_MIN US_OA	post_hoc_cross_outcome	135	-0.252	[-0.403, -0.086]	0.0032	0.0219	0.0198
NET07	DELTA_OC_MIN US_OA	post_hoc_cross_outcome	135	0.227	[0.061, 0.382]	0.0082	0.0219	0.0474
NET08	DELTA_OC_MIN US_OA	same_construct_external_main_effect	135	0.112	[-0.058, 0.276]	0.1951	0.1951	0.6334

For BAI-derived signatures, testing against MSPSS was a cross-outcome post hoc analysis.

S6. Machine-readable files

Supplementary_Tables.xlsx mirrors the principal audit tables. The tables/ directory contains the complete row-level aggregate outputs, including all 576 local correlations, 1,920 local moderation models, 750 spatial aggregation models, 30 selection-aware signature combinations, 112 external signature interactions, 700 external target interactions, and the external main-effect families. The manifests/ directory records seeds, exchangeability schemes, software versions, source hashes, and archive hashes.